Popliteus / calf deceleration prevention protocol

Post-badminton flare rebuild · hypermobility-safe · 3-phase progressive (~10 wks) · knee rehab phase

Muscle group

Calves · Popliteus · Glutes

Why this file exists — and what runs alongside it

Rested 2 months to full resolution after a flare here — this rebuilds the load-capacity tier that failed (deceleration/lunge overload), not muscle mass. Run the 5-test baseline battery first to set Phase 1 length.
PREREQUISITE: the lower-body mobility protocol is the pre/post wrapper for every session in this file. Its Ankle CARs, clamshells and banded TKEs prime exactly what this file loads — never skip on reactive days.

A — Phase 1: Reintroduce (weeks 1–2 · bilateral only · submaximal)

1	Straight-knee calf isometric hold Gastrocnemius — landing shock absorber Re-establish tolerance before loading · knee stays soft	5 × 30–45s Bodyweight

↕ then immediately

2	Popliteus banded internal rotation (non-stance foot draws behind support leg · slow controlled return) Popliteus — resists tibial rotation & anterior translation	2–3 × 12 Band
3	Bent-knee calf isometric hold Soleus — primary plantarflexor with knee bent = exact lunge/landing position; deceleration-specific Straight set — Phase 1 hip ER dropped (mobility protocol covers it)	5 × 30–45s Bodyweight

B — Phase 2: Strength base (weeks ~2–6 · length set by baseline results)

4	Standing calf raise (Smith · optional deficit) (2s up / 3s down · soft knee · 70–80% range only) Gastrocnemius + soleus synergist — the failed capacity tier Progress to single-leg late-phase — ankle eccentric only, knee stays static	3–4 × 8–12 Smith

↕ then immediately

5	Popliteus step task (stance foot on 5–15cm step · knee 20–40° · stays planted) (free leg: behind+across → fwd → behind+inside → fwd → out to side) Popliteus — stance leg RESISTS rotation; free leg supplies the torque Controlled tempo to learn the sequence · then swap legs	3 × 5 cycles/leg Low step
6	Seated calf raise (plate / DB on knees) (knee ~80° · 2s up / 3s down · soft lockout) Soleus — non-negotiable slot; this is your deceleration muscle	3–4 × 10–15 Plate / DB

↕ then immediately

7	Standing banded hip ER / lateral step Glute med/max — controls valgus & tibial-rotation overflow LOADED dose — activation dose is covered pre-session by the mobility protocol	2–3 × 12–15 Band

C — Phase 3: Reactive / deceleration (weeks ~6–10 · enter only once Phase 2 is pain-free)

8	Bilateral pogo hops (low amplitude · ankle-stiff · done fresh, before calf loading) Gastroc / soleus / Achilles — reactive energy-storage capacity	3–4 × 20–30s Bodyweight

↕ then immediately

9	Popliteus step task — at speed (same 5-position sequence · quick loading-unloading, max speed) (form holds first · then progress onto unstable surface) Popliteus — rotatory control under speed, not just under load	3 × 5 cycles/leg Low step
10	Sub-max lunge landing (step-in → hold) (step in slow · no absorb · hold 2–3s at moderate depth) GATED: add the absorb/landing phase only once knee clears SL eccentric Straight set — protect from fatigue stacking (superset Rule 4)	3 × 5/leg Bodyweight

Return-to-badminton gate

Footwork drills → half-speed rallies → full session — never straight into a heavy one

Every session — constraints across all phases

- Lower-body mobility protocol runs pre + post — every session in this file, no exceptions
- Soft lockout throughout; ~70–80% ROM only — no end-range loaded stretch
- No passive stretching — active mobility / isometric holds only
- Single-leg eccentric restriction is KNEE-only — ankle/calf eccentrics already live in mobility protocol
- Deep loaded knee/hip flexion stays out — Phase 3 stays bodyweight, low-amplitude
- Any symptom reproduction during testing or training overrides phase length
- Hip abductor/adductor machine: still deferred until knee stability improves

Phase length & gating logic

Entry gate: near norms + clean control + no pain → Phase 1 ~2wk. Deficit / asymmetry >10% / poor control / pain → extend base to ~4–5wk; re-test before advancing. Phase 3 entry requires Phase 2 pain-free AND strong.
Re-test the baseline battery every 2–3 weeks.
Integrates into the split: Phases 1–2 → Monday (lower) · Phase 3 reactive → Friday (cardio).

Sets per session, by phase

Phase 1: ~12–13 sets (1 pair + 1 straight set · isometric)
Phase 2: ~11–14 sets (2 pairs · strength + control)
Phase 3: ~9–10 sets (1 pair + 1 straight set · reactive)

Alternative exercises — if Smith machine occupied

# Primary exercise	Alternative 1	Alternative 2
4 Standing calf raise (Smith)	DB standing calf raise (hold rack)	Deficit calf raise (step edge)